

Hybrid Neighborhood Evaluation in Tabu Search and Simulated Annealing for the Permutation Flow Shop Problem

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Outline

- 1 Introduction
- 2 Elementary Neighborhoods
- 3 Hybridization Strategies
- 4 Algorithm
- 5 Experiments
- 6 Conclusion

Introduction

- **Permutation Flow Shop Problem (PFSP):** NP-hard for $m \geq 3$. Metaheuristics rely on a chosen *neighborhood* to explore the solution space.
- Single-neighborhood local search has systematic blind spots:
 - **Adjacent swap:** cheap, but only short-range moves.
 - **Dynasearch:** expensive per iteration.
 - **Motzkin arcs:** cover specific structural patterns only.
- **Idea:** combine multiple elementary neighborhoods into a hybrid scheme that adapts to the current state of the search.
- This work: five hybridization strategies, two metaheuristic frameworks (TS, SA), four elementary neighborhoods — on Taillard benchmarks.

Problem and Elementary Neighborhoods

PFSP makespan: minimize $C_{\max}(\pi) = C_{m-1,n-1}(\pi)$ over $\pi \in S_n$.

Four elementary neighborhoods:

Symbol	Move	Size / cost
N_{adj}	swap $\pi_i \leftrightarrow \pi_{i+1}$	$n-1$, $O(1)$ per Δ
N_{fib}	reinsert at $i + F_k$ (Fibonacci jumps)	$O(n \log n)$, selective long-range
N_{dyn}	DP-optimal composite swap	$O(n^2)$ per application
N_{motz}	non-crossing arc moves	$M_n \sim 3^n/n^{3/2}$, filtered

All neighborhoods accelerated by the Smutnicki block-property and the NPI filter.

Five Hybridization Strategies

Let $\mathcal{N} = \{N_1, \dots, N_K\}$. At iteration t select one $N_{s(t)}$:

- **FC — Fixed Cycle:** $s(t) = (t \bmod K) + 1$. Baseline.
- **PW — Probability-Weighted:** $s(t) \sim \text{Categorical}(w)$ with tuned weights.
- **PA — Performance-Adaptive:** ε -greedy on EMA of past improvements per neighborhood.
- **BK — Best-of- K :** evaluate all K neighborhoods, take the best move. Best quality, $K \times$ cost.
- **DP — Dynamic Programming:** finite-horizon MDP over (π, t) ; backward induction with horizon $H = 5$. Only feasible for $n \leq 50$.

Performance-Adaptive Strategy — in Detail

EMA of improvement per neighborhood:

$$\bar{\delta}_k^{(t+1)} = \gamma \bar{\delta}_k^{(t)} + (1 - \gamma) \delta_k^{(t)}, \quad \gamma = 0.9$$

where $\delta_k^{(t)}$ is the actual $\Delta C_{\max}$ gained by N_k at step t (0 if not selected).

Selection rule (ε -greedy):

$$s(t) = \begin{cases} \arg \max_k \bar{\delta}_k^{(t)} & \text{with probability } 1 - \varepsilon, \\ \text{uniform random } \in \mathcal{N} & \text{otherwise.} \end{cases}$$

Why it works: matches **Best-of- K** quality at $1/K$ of the per-iteration cost — it learns the best neighborhood for the current basin instead of trying all every step.

Hybrid Tabu Search (HTS)

input : initial π_0 , time budget $T_{\max}$

output: best permutation π^*

$\pi \leftarrow \pi_0$; $\pi^* \leftarrow \pi$; $\mathcal{T} \leftarrow \emptyset$; $\mathcal{M} \leftarrow \text{MushroomList}(k=10)$;

while $time < T_{\max}$ **do**

 select $N_{s(t)}$ by hybridization strategy;

$(\pi', m) \leftarrow$ best non-tabu move in $N_{s(t)}(\pi)$ (or aspiration if it beats π^*);

if $m = \emptyset$ **then**

$\pi \leftarrow \mathcal{M}.\text{perturb}()$ *// double-bridge from elite pool*;

else

$\pi \leftarrow \pi'$; $\mathcal{T} \leftarrow \mathcal{T} \cup \{m\}$;

 update π^* , $\mathcal{M}$, and EMAs $\bar{\delta}_k$;

return π^* ;

Experimental Setup

- **Benchmarks:** Taillard instances, $n \in \{20, 50, 100, 200\}$, $m \in \{5, 10, 20\}$, 10 instances per class.
- **Time budget:** $T_{\max} = 60\,000$ ms.
- **Metaheuristics:** Tabu Search ($\tau = n/5$, aspiration, MushroomList $k=10$) and Simulated Annealing (exp. cooling, $\alpha = 0.995$, adaptive rehear).
- **Quantum variants:** D-Wave Advantage 4.1, 5,000 reads, windowed decomposition for $n > 19$ (window 19, overlap 9).
- **Metric:** mean relative percentage deviation (RPD) to best-known $C_{\max}^*$:

$$\text{RPD} = \frac{C_{\max} - C_{\max}^*}{C_{\max}^*} \times 100\%.$$

Results — Mean RPD across Taillard Instances

Tabu Search					Simulated Annealing				
Strategy	$n=20$	50	100	200	Strategy	$n=20$	50	100	200
Adjacent (single)	1.82	2.41	3.18	4.02	Adjacent (single)	1.91	2.55	3.31	4.21
Dynasearch (single)	1.64	2.22	2.97	3.79	Dynasearch (single)	1.75	2.37	3.08	3.91
FC	1.71	2.18	2.91	3.71	FC	1.79	2.29	2.98	3.78
PW	1.58	2.09	2.83	3.62	PW	1.66	2.18	2.87	3.66
PA	1.44	1.97	2.71	3.49	PA	1.53	2.04	2.76	3.54
BK	1.47	2.01	2.74	3.52	BK	1.56	2.08	2.79	3.57

PA wins in every cell, matching BK quality at $1/K$ of the cost; quantum-assisted variants gain a further 0.08–0.15 pp at $n = 200$.

Conclusion and Future Work

Summary:

- Hybrid neighborhoods consistently beat single-neighborhood baselines in both TS and SA.
- **Performance-Adaptive (PA)** is the sweet spot: best-of- K quality at $1/K$ of the per-iteration cost.
- Combining Motzkin and Fibonacci alongside Adjacent and Dynasearch matters most for large instances ($n = 200$).
- Quantum-assisted neighborhoods give an extra 0.08–0.15 pp at $n = 200$.

Future work:

- Multi-armed bandit formulations for neighborhood selection (UCB, Thompson sampling).
- Quantum-native hybrid strategies exploiting D-Wave's native graph structure.
- Joint hybridization across TS and SA (portfolio approach).